GOOD COORDINATES IN THE FOURIER ENTROPY-INFLUENCE CONJECTURE

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ABSTRACT. For a Boolean function f let $E_i(f) = H(\mathbf{1}[i \in U] \mid U \setminus \{i\})$, where U is drawn from the spectral distribution $\hat{f}^2$. The Fourier entropy-influence conjecture holds if and only if the quantity $\mathsf{G} = \sup_f \min_i E_i(f)/\mathrm{Inf}_i(f)$ is finite, and then its constant lies between G and $\mathsf{G} + 2/\ln 2$. We study G directly. A composition identity computes E_i and Inf_i for iterated block compositions $g(h, \dots, h)$ exactly from the biased spectrum of g, and towers of transitive monotone gates give $\mathsf{G}_{\mathrm{mon}} \geq 4.50642846$, where $\mathsf{G}_{\mathrm{mon}}$ is the supremum over monotone functions. On the other side we show that every width-w DNF has a coordinate with $E_i/\mathrm{Inf}_i \leq 2 + \log_2 e + \log_2(2^{w+1} - 1)$, with no monotonicity or read restriction, and that every relevant leaf of a read-once AND/OR formula has $E_i/\mathrm{Inf}_i < \pi/\ln 2$, which is sharp. Finally we show that the coordinate of largest influence is not a valid selection rule, and that all but a vanishing fraction of the coordinates of a monotone function can have E_i/Inf_i bounded below by any constant. Nor can the ratios be averaged with weights Inf_i^{a+1} for any $a \neq 0$.

1. INTRODUCTION

Let $f : \{-1,1\}^n \to \{-1,1\}$ have Fourier expansion $f = \sum_S \hat{f}(S)\chi_S$, and let $U \subseteq [n]$ be drawn with $\Pr[U = S] = \hat{f}(S)^2$. The spectral entropy of f is $H(f) = H(U)$ and the influence of i is $\mathrm{Inf}_i(f) = \Pr[i \in U]$. The Fourier entropy-influence (FEI) conjecture of Friedgut and Kalai [5] asserts that $H(f) \leq C \sum_i \mathrm{Inf}_i(f)$ for a universal constant C; let C^* be the least such constant, $C^* = \infty$ if there is none. The conjecture is known for several classes [19, 4, 10, 26, 20], among them the linear threshold functions [23], and with a log-type loss in general [7, 10]; Kelman, Kindler, Lifshitz, Minzer and Safra [9] prove a weaker form, that the spectrum of a Boolean function of constant variance is concentrated on $2^{O(\mathrm{I}(f)\log \mathrm{I}(f))}$ characters. The largest known lower bound is $C^* > 6.5218$, from an explicit balanced monotone function on 18 variables [16] through the amplification $C^* \geq H(g)/(\mathrm{I}(g) - 1)$ of O'Donnell and Tan [18]; it improves the bounds 6.278 [18], 6.4548 [8], 6.4901 and 6.5143 [14, 15] (the records are collected in [25]).

González, MacManus and Pereyra [6] reduce the conjecture to a statement about one coordinate: it is enough that every Boolean function has a coordinate i at which the entropy drop $H(f) - \frac{1}{2}H(f_{i\to 1}) - \frac{1}{2}H(f_{i\to -1})$ is at most $K\mathrm{Inf}_i(f)$. They also observe that the choice of coordinate matters: for their δ-tribes functions a stronger form of the inequality fails at one coordinate and holds, with constant 38, at another [6, Section 5]. In [22] it is shown that the reduction is lossless up to an additive $2/\ln 2$, in the following form. Write

$$E_i(f) = H\big(\mathbf{1}[i \in U] \,\big|\, U \setminus \{i\}\big), \qquad \mathsf{G}(f) = \min_{i:\, \mathrm{Inf}_i(f) > 0} \frac{E_i(f)}{\mathrm{Inf}_i(f)}, \qquad \mathsf{G} = \sup_f \mathsf{G}(f),$$

the supremum over all nonconstant Boolean functions on any number of variables. Then

$$\mathsf{G} \leq C^* \leq \mathsf{G} + \frac{2}{\ln 2}.$$

So the conjecture is equivalent to $\mathsf{G} < \infty$, and any proof of it that proceeds by choosing one good coordinate at a time needs a selection rule whose ratio E_i/Inf_i is bounded. This paper

studies G itself: how large it must be, which classes have a coordinate with bounded ratio, and which natural selection rules fail.

Results. Section 2 gives an exact formula for E_i and Inf_i under block composition $f = g(h, \dots, h)$ (Theorem 2.1). The formula carries four numbers per leaf through each level of a composition, so towers of depth ten or more on 10^{12} and more variables are computed in milliseconds and to any precision.

Section 3 uses it to bound $\mathsf{G}_{\mathrm{mon}}$, the supremum of $\mathsf{G}(f)$ over monotone f, from below. When every gate is transitive on its inputs, every leaf of the tower has the same ratio, so the ratio of one leaf is $\mathsf{G}(f)$. Towers of the perfect matching function of $K_{3,3}$ and its dual already give $\mathsf{G}_{\mathrm{mon}} > 4.505$; mixed words of six gate types give

$$\mathsf{G} \geq \mathsf{G}_{\mathrm{mon}} \geq 4.5064284654452331779\dots$$

This is below the bound $C^* > 6.5218$, as it must be: towers are built to make one coordinate's ratio large, and the spectral entropy of these functions is far from extremal. The point of the bound is the other one: any proof of FEI through a good coordinate must allow ratio at least 4.506 already for monotone functions.

Section 4 bounds $\mathsf{G}(f)$ from above for classes. Every coordinate satisfies $E_i/\mathrm{Inf}_i \leq 2 + \log_2 e + \log_2(1 + \mathrm{Var}(f)/\mathrm{Inf}_i)$, and from this every width-w DNF or CNF has a coordinate with ratio at most $2 + \log_2 e + \log_2(2^{w+1} - 1)$. For read-once AND/OR formulas every relevant leaf has ratio below $\pi/\ln 2$, and the constant is attained in the limit.

Section 5 gives three negative results. The coordinate of largest influence can have unbounded ratio, even for monotone functions. In a monotone function built from a middle-layer slice all but k of $\binom{k}{\lfloor k/2 \rfloor} + k$ coordinates can have ratio above any prescribed constant. And the weighted inequalities $\sum_i \mathrm{Inf}_i^a E_i \leq C \sum_i \mathrm{Inf}_i^{a+1}$, which would give a selection rule by averaging, fail for every $a \neq 0$. The case $a = 0$ is the conjecture itself.

All numerical values in this paper are reproduced by the scripts in the accompanying repository, in double precision and with 40-digit arithmetic.

2. Composition

2.1. **Biased expansion of the gate.** Let $g : \{-1,1\}^k \to \{-1,1\}$ and $\mu \in (-1,1)$. Put $\sigma = \sqrt{1-\mu^2}$ and $\phi_\mu(y) = (y-\mu)/\sigma$. If $y_1, \dots, y_k$ are independent with mean μ, the products $\prod_{l \in T} \phi_\mu(y_l)$ are orthonormal, and

$$g(y) = \sum_{T \subseteq [k]} \hat{g}_\mu(T) \prod_{l \in T} \phi_\mu(y_l), \qquad \hat{g}_\mu(T) = \mathbb{E}\Big[g(y) \prod_{l \in T} \phi_\mu(y_l)\Big], \qquad \sum_T \hat{g}_\mu(T)^2 = \mathbb{E}[g^2] = 1.$$

In particular $\hat{g}_\mu(\emptyset) = \mathbb{E}[g(y)]$.

2.2. **The spectral sample of a composition.** Let $h : \{-1,1\}^m \to \{-1,1\}$ have mean $\mu = \hat{h}(\emptyset) \in (-1,1)$, and let $f(x^1, \dots, x^k) = g(h(x^1), \dots, h(x^k))$ on k disjoint blocks of m variables. Since $\phi_\mu(h) = \sigma^{-1} \sum_{A \neq \emptyset} \hat{h}(A) \chi_A$, expanding g gives

$$\hat{f}\Big(\bigcup_{l \in T} A_l\Big) = \hat{g}_\mu(T) \prod_{l \in T} \frac{\hat{h}(A_l)}{\sigma} \qquad (A_l \neq \emptyset \text{ a subset of block } l),$$

and every Fourier coefficient of f has exactly one such representation. Since $\sum_{A \neq \emptyset} \hat{h}(A)^2 = 1 - \mu^2 = \sigma^2$, the spectral sample of f is the following two-stage draw: choose $T \subseteq [k]$ with probability $\hat{g}_\mu(T)^2$, then, independently for each $l \in T$, choose a nonempty A_l in block l with probability $\hat{h}(A_l)^2/\sigma^2$, and set $U = \bigcup_{l \in T} A_l$.

2.3. **The leaf state.** Fix a coordinate i of h. Write $m(a,b) = (a+b)\,h_2\big(b/(a+b)\big)$ for $a,b \ge 0$, where $h_2(p) = -p\log_2 p - (1-p)\log_2(1-p)$ and $m(0,0) = 0$; this is the unnormalised entropy of a two-point mass. Grouping the spectral sets of h according to $B = U \setminus \{i\}$,

$$E_i(h) = \sum_{B \not\ni i} m\big(\hat{h}(B)^2, \hat{h}(B \cup \{i\})^2\big) = e(h) + m\big(\mu^2, \hat{h}(\{i\})^2\big), \tag{1}$$

where $e(h)$ is the sum over $B \neq \emptyset$. We call $(\mu,\ \hat{h}(\{i\})^2,\ e(h),\ \mathrm{Inf}_i(h))$ the state of the leaf i in h.

Theorem 2.1. *Let $f = g(h,\dots,h)$ as above, let i be a coordinate of h placed in block j, and let (μ, P, e, I) be its state in h. Write $\hat{g}$ for $\hat{g}_\mu$ and $W_j = \sum_{T \ni j} \hat{g}(T)^2$, the μ-biased influence of j on g. Then the state (μ', P', e', I') of i in f is*

$$\mu' = \hat{g}(\emptyset), \qquad P' = \hat{g}(\{j\})^2\,\frac{P}{\sigma^2},$$
$$I' = W_j\,\frac{I}{\sigma^2}, \qquad e' = W_j\,\frac{e}{\sigma^2} + \sum_{\substack{R \neq \emptyset \\ j \notin R}} m\Big(\hat{g}(R)^2,\ \hat{g}(R \cup \{j\})^2\,\frac{P}{\sigma^2}\Big),$$

and $E_i(f) = e' + m(\mu'^2, P')$.

Proof. $\mu' = \mathbb{E} f = \hat{g}(\emptyset)$, and $P' = \hat{f}(\{i\})^2$ is the case $T = \{j\}$, $A_j = \{i\}$ of the product formula. For I': $i \in U$ exactly when $j \in T$ and $i \in A_j$, and these events are independent with probabilities W_j and $\sum_{A \ni i} \hat{h}(A)^2/\sigma^2 = I/\sigma^2$.

For $E_i(f)$ we condition on $U \setminus \{i\}$, which determines $R = T \setminus \{j\}$, the sets A_l for $l \neq j$, and $B = A_j \setminus \{i\}$ when $j \in T$. Whether $i \in U$ depends only on what happens in block j.

If $B \neq \emptyset$ then $j \in T$, and given B the event $i \in A_j$ has the conditional law it has in the normalised nonempty spectrum of h; the blocks $l \neq j$ are independent of it. Summing over $T \ni j$ and over nonempty B gives $W_j\,e/\sigma^2$.

If $B = \emptyset$ then either $j \notin T$, so $T = R$ and $i \notin U$, with probability $\hat{g}(R)^2$, or $T = R \cup \{j\}$ and $A_j = \{i\}$, with probability $\hat{g}(R \cup \{j\})^2 P/\sigma^2$. The outcomes in the other blocks factor out of both alternatives with the same weight, so this part contributes $\sum_{R \not\ni j} m(\hat{g}(R)^2, \hat{g}(R \cup \{j\})^2 P/\sigma^2)$. The term $R = \emptyset$ is the one in which $U \setminus \{i\} = \emptyset$; it equals $m(\mu'^2, P')$, and the remaining terms together with $W_j e/\sigma^2$ form e', the part of $E_i(f)$ over nonempty $U \setminus \{i\}$. □

A single variable x_i has state $(0,1,0,1)$. Iterating Theorem 2.1 computes the state of any leaf of an iterated composition $g_d(\dots g_2(g_1(x,\dots),\dots)\dots)$, and the cost of a level is one biased transform of the gate, $O(k2^k)$, independent of the number of variables below it. We checked the formula against brute-force Fourier transforms on all 444 leaf positions of depth-two compositions of gates on three and four variables, with largest discrepancy 3.3×10^{-16}.

Remark 2.2. If g is transitive on its inputs, so is the composition on its leaves, and every leaf has the same state. For such towers the ratio of one leaf is $\mathsf{G}(f)$ and also $\max_i E_i/\mathrm{Inf}_i$; a lower bound on G then needs no search over coordinates. Without transitivity the formula still applies to each leaf separately, with W_j and $\hat{g}(\{j\})$ taken at that leaf's position.

Remark 2.3. Keller, Mossel and Schlank [10, Claim 4.1] bound $H(f)$ above by $\sum_i h_2(\mathrm{Inf}_i)$ using subadditivity of entropy over the indicators $\mathbf{1}[i \in U]$. The quantity E_i is the conditional counterpart: it measures how much of the uncertainty of $\mathbf{1}[i \in U]$ remains once the rest of U is known.

3. Towers of monotone gates

3.1. **Gates and convention.** We write a gate as a monotone function of $z \in \{0,1\}^k$ and pass to ± 1 values by $x = -1$ when $z = 1$ and output -1 when the gate is true. The gates used are:

- M, the perfect-matching function of $K_{3,3}$: z indexes the nine edges and $M(z) = 1$ when the chosen edges contain a perfect matching; $D(z) = 1 - M(\mathbf{1} - z)$ is its dual;
- P, the perfect-matching function of $K_{4,4}$ on sixteen edges, and its dual P^*;
- the tribes functions $T(w, s)$, the OR of s disjoint ANDs of w variables each, and their duals $T(w, s)^*$; we also write $Q_2 = T(2, 2)^*$, and R^* for the dual of the function on the six edges of K_4 that is true when two opposite edges are both present (a copy of $T(2, 3)$).

Each gate is monotone and its automorphism group is transitive on its inputs. A word $g_1 g_2 \cdots g_d$ denotes the iterated composition in which g_1 is applied to disjoint blocks of variables, g_2 to disjoint blocks of copies of the result, and so on; a word is read from the bottom gate to the top gate. Compositions of monotone functions are monotone, and since every gate is transitive the composition is transitive on its leaves, so $\mathsf{G}(f) = E_i(f)/\mathrm{Inf}_i(f)$ for any leaf i. That ratio is computed by iterating Theorem 2.1 from the state $(0, 1, 0, 1)$ of a single variable.

Proposition 3.1. *For the words in Table 1 the iterated compositions are monotone Boolean functions with $\mathsf{G}(f)$ equal to the value shown, to the number of digits shown. In particular*

$$\mathsf{G}_{\mathrm{mon}} \geq 4.50642846544523317790928639\ldots$$

word (bottom to top)	variables	$\mathsf{G}(f)$
$M\,D\,D\,D\,D\,T(2,5)^*$	5.9×10^5	4.505850062076900302064374353 06
$M\,D\,D\,M\,M\,D\,D$	4.8×10^6	4.505049967497627926957101334 17
$M\,D\,D\,D\,D\,D\,M\,M$	4.3×10^7	4.506379879467710610635356813 98
$D\,M^{12}$	2.5×10^{12}	4.506428305525435411086248568 79
$M\,D^4\,M\,P\,P\,P\,Q_2\,R^*\,T(4,3)^*\,T(2,6)\,T(3,5)$	1.1×10^{14}	4.506400645515568435507173670 45
$D\,M^8\,P^*\,M\,M\,T(3,5)$	7.5×10^{12}	4.506428462939128775313109412 80
$M\,D^8\,P\,D\,D\,T(4,3)^*\,T(3,5)\,T(3,5)$	1.4×10^{15}	4.506428465445233177909286392 45

Table 1. Lower bounds for $\mathsf{G}_{\mathrm{mon}}$ from towers. Values to 30 digits are computed with 40-digit arithmetic.

Proof. Each value is the output of Theorem 2.1 iterated along the word, with the biased coefficients of each gate computed from its truth table. The computation is exact up to rounding, and was carried out with 40 significant digits; the digits shown are stable under a change to 60. For $D\,M^{12}$ and the last two rows the iteration was also run in 60-digit interval arithmetic with outward rounding; each enclosure has width below 10^{-48} and contains the value shown. Theorem 2.1 itself was checked against brute-force Fourier transforms as described at the end of Section 2. □

Remark 3.2. Double precision is not enough to rank long towers. For the last word of Table 1 the double-precision iteration returns 4.506428469267, an error of 3.8×10^{-9}, which is larger than the gap of 2.5×10^{-9} between the last two rows. The ordering of the table is that of the 40-digit values.

Remark 3.3 (Where the value comes from). For M the mean map $\mu \mapsto \hat{M}_\mu(\emptyset)$ has a fixed point at $\mu^* = -0.0349185099940\ldots$, where its derivative is 2.0077, so the fixed point is repelling. The bottom gate D produces the mean -0.03515625, close to μ^*, and a run of M gates then keeps the mean near μ^* for several levels before it escapes; the top gates spend what is left. If the

mean could be held at μ^* for ever, starting from the state produced by D, the ratio would tend to 4.5064554626637. This is a heuristic target rather than a bound: no finite tower holds the mean at a repelling fixed point, the limit depends on the starting state (4.5064428 from the state after $D\,M$, 4.5064353 after $D\,M\,M$), and we do not know whether $\mathsf{G}_{\mathrm{mon}}$ exceeds it. The same heuristic for other gates gives larger targets, 4.5168 for $T(2,2)^*$ pinned at its fixed mean 0.236068 and 4.5107 for $T(4,3)^*$, but towers that bring the mean close to these fixed points carry a smaller leaf ratio, and none of depth at most four below the pinned gate exceeds 4.504.

4. Classes with a good coordinate

Throughout this section f is Boolean, i is a coordinate with $\delta = \mathrm{Inf}_i(f) > 0$, and for $T \not\ni i$ we put $p_T = \hat{f}(T)^2$ and $q_T = \hat{f}(T \cup \{i\})^2$. Conditioning on $U \setminus \{i\} = T$ gives

$$E_i(f) = \sum_{T \not\ni i} m(p_T, q_T), \qquad \mathrm{Inf}_i(f) = \sum_{T \not\ni i} q_T. \tag{2}$$

The function m is concave, increasing in each argument and homogeneous of degree one, hence superadditive: $\sum_T m(a_T, b_T) \le m(\sum_T a_T, \sum_T b_T)$. Write $g = \mathbb{E}_i f$ and $h = \mathrm{D}_i f$, so that $f = g + x_i h$, $p_T = \hat{g}(T)^2$ and $q_T = \hat{h}(T)^2$; since h takes values in $\{-1, 0, 1\}$, $\sum_T q_T = \mathbb{E}h^2 = \delta$.

Lemma 4.1. *For every Boolean f and every coordinate i with $\delta = \mathrm{Inf}_i(f) > 0$,*

$$\frac{E_i(f)}{\delta} \le 2 + \log_2 e + \log_2\Big(1 + \frac{\mathrm{Var}(f)}{\delta}\Big).$$

Proof. For $t \in [0,1]$ one has $h_2(t) \le 2\sqrt{t(1-t)}$, so $m(a,b) \le 2\sqrt{ab}$. The term $T = \emptyset$ is therefore at most $2\sqrt{q_\emptyset} = 2|\mathbb{E}h| \le 2\delta$. For the others, $\sum_{T \ne \emptyset} p_T = \mathrm{Var}(g) \le \mathrm{Var}(f) = V$ and $\sum_{T \ne \emptyset} q_T \le \delta$, so by superadditivity and monotonicity

$$\sum_{T \ne \emptyset} m(p_T, q_T) \le m(V, \delta) = V \log_2\Big(1 + \frac{\delta}{V}\Big) + \delta \log_2\Big(1 + \frac{V}{\delta}\Big) \le \frac{\delta}{\ln 2} + \delta \log_2\Big(1 + \frac{V}{\delta}\Big). \qquad \square$$

So a coordinate whose influence is a fixed fraction of the variance is good. Width supplies one.

Theorem 4.2. *Let f be a nonconstant function computed by a DNF in which every term has at most w literals. Then f has a coordinate with*

$$\frac{E_i(f)}{\mathrm{Inf}_i(f)} \le 2 + \log_2 e + \log_2\big(2^{w+1} - 1\big) < w + 3 + \log_2 e.$$

The same holds for CNFs of width w. No monotonicity and no bound on the number of occurrences of a variable is assumed.

Proof. Since f is not constant, some term is satisfiable and nonempty; let its literals be $\ell_1, \dots, \ell_k$, $k \le w$, on coordinates $j_1, \dots, j_k$. From every input x with $f(x)$ false, set $\ell_1, \dots, \ell_k$ true one at a time. The last input satisfies the term, so some step crosses an edge of the cube along which f changes. An edge in direction j_t is reached at step t from at most 2^{t-1} starting inputs, those that differ from its lower end only in $j_1, \dots, j_{t-1}$. The number of such edges in direction j is $2^{n-1}\mathrm{Inf}_j(f)$. Hence

$$\Pr[f \text{ false}] \le \frac{1}{2} \sum_{t=1}^{k} 2^{t-1} \mathrm{Inf}_{j_t}(f) \le \frac{2^k - 1}{2} \max_t \mathrm{Inf}_{j_t}(f).$$

Take i the maximiser. With $V = \mathrm{Var}(f) = 4 \Pr[f \text{ false}] \Pr[f \text{ true}]$ this gives $V/\mathrm{Inf}_i(f) \le 2(2^k - 1)$, and Lemma 4.1 gives the bound. For a CNF apply this to $\neg f$, which has the same E_i and Inf_i. $\square$

Width is preserved by restriction, so the induction of [22] turns Theorem 4.2 into $H(f) \le (w + 3 + 3\log_2 e)\,\mathrm{I}(f)$ for width-w DNFs and CNFs. The argument has the same shape as the coordinate choice of [6, Section 5.2]: a variable inside a short term sees a positive fraction of the variance. For comparison, Han [7] proves

$$H(f) \le \Big(2 + \frac{3}{\ln 2}\Big)\mathrm{I}(f) + \sum_i \mathrm{Inf}_i(f) \log_2 \frac{1}{\mathrm{Inf}_i(f)}$$

for every f, Li and Han [11] lower the constant $2 + 3/\ln 2$ to the sharp value $1/\ln 2$, and Arunachalam, Chakraborty, Koucký, Saurabh and de Wolf [1] bound $H(f)$ by twice the average unambiguous parity-certificate complexity.

4.1. **Read-once formulas.** For read-once formulas a much stronger statement holds: every relevant leaf is good, with a sharp constant.

Theorem 4.3. *Let f be a read-once formula over AND and OR with literals at the leaves. Every leaf i with $\mathrm{Inf}_i(f) > 0$ satisfies*

$$\frac{E_i(f)}{\mathrm{Inf}_i(f)} < \frac{\pi}{\ln 2} = 4.5323601418\ldots$$

and the constant $\pi/\ln 2$ cannot be lowered: there are monotone read-once formulas and leaves with ratio arbitrarily close to it.

Negating an input or the output changes neither E_i nor Inf_i, so we may assume f monotone. Let the gates on the path from leaf i to the root be numbered $1, \ldots, L$ upwards. At gate l the other children form a read-once formula on a block of fresh variables; let $Z_l \in \{0,1\}$ be the event that they let the path child through (all true at an AND, all false at an OR), $p_l = \mathbb{E}Z_l$, and $c_l \in \{-1, 1\}$ the output when they block it. We may assume $0 < p_l < 1$: a gate with $p_l = 1$ can be deleted and one with $p_l = 0$ makes i irrelevant. Then $f_0 = x_i$, $f_l = c_l(1 - Z_l) + f_{l-1}Z_l$ and $f = f_L$. Put $P_0 = 1$, $P_l = p_1 \cdots p_l$ and $\delta = P_L = \mathrm{Inf}_i(f)$.

Lemma 4.4. *Let $\Phi(u) = m(u^2, 1)$. With $R_r = \sum_{l<r} c_l(1 - p_l)/P_l - c_r/P_{r-1}$ and $R_\infty = \sum_{l \le L} c_l(1 - p_l)/P_l$,*

$$\frac{E_i(f)}{\delta} = \sum_{r=1}^{L} (P_{r-1} - P_r)\,\Phi(R_r) + \delta\,\Phi(R_\infty).$$

Proof. Unrolling the recursion, $h = \mathrm{D}_i f = \prod_l Z_l$ and $g = \mathbb{E}_i f = \sum_l c_l(1 - Z_l)\prod_{j>l} Z_j$. A frequency $S \not\ni i$ splits into block parts S_l, and $\hat{h}(S) = \prod_l \hat{Z}_l(S_l)$ with $\hat{Z}_l(\emptyset) = p_l$. If r is the first l with $S_l \ne \emptyset$, only the terms $l \le r$ of g contribute, and $\hat{g}(S) = R_r\hat{h}(S)$; if all S_l are empty, $\hat{g}(S) = R_\infty\hat{h}(S)$. By homogeneity $m(p_S, q_S) = q_S\Phi(\hat{g}(S)/\hat{h}(S))$ when $\hat{h}(S) \ne 0$, and $m(p_S, 0) = 0$. Under the law q_S/δ the blocks are independent, block l is empty with probability p_l, so the first nonempty block is r with probability $P_{r-1} - P_r$ and none is nonempty with probability δ. Now apply (2). □

Proof of Theorem 4.3. Let $a_l \le b_l$ be the probabilities that f_l is true when x_i is false, respectively true, and $d_l = b_l - a_l = P_l$. Put $\alpha = a/d$, $\beta = (1 - b)/d$, so $s = 1 + \alpha + \beta = 1/d$. An AND gate keeps α and raises β; an OR gate keeps β and raises α. In terms of the values (α, β) before gate r, $\sum_{l<r} c_l(1 - p_l)/P_l = \beta - \alpha$ (with $c = -1$ for true), so $|R_r| = 1 + 2\alpha$ at an AND and $|R_r| = 1 + 2\beta$ at an OR, and $|R_\infty| = |\beta - \alpha|$ at the end. With $\psi(t) = \Phi(2t + 1)$, the charge of an AND step is

$$(P_{r-1} - P_r)\,\psi(\alpha) = \int_\beta^{\beta'} \frac{\psi(\alpha)}{(1 + \alpha + \beta)^2}\,d\beta,$$

and symmetrically at an OR. The function Φ is increasing on $[0,\infty)$, and for $u \ge 1$

$$\Phi''(u)\ln 2 = 2\ln(1+u^{-2}) - \frac{4}{1+u^2} < 0,$$

since $\ln(1+x) < 2x/(1+x)$ for $0 < x \le 1$. So ψ is increasing and strictly concave on $[0,\infty)$. For $\alpha \ge \beta$ let

$$W(\alpha) = \int_\alpha^\infty \frac{2\psi(t)}{(1+2t)^2}\,dt, \qquad V(\alpha,\beta) = \psi(\alpha)\Big[\frac{1}{s} - \frac{1}{1+2\alpha}\Big] + W(\alpha),$$

and $V(\alpha,\beta) = V(\beta,\alpha)$ for $\alpha < \beta$; V is continuous. For $\alpha \ge \beta$, $-V_\beta = \psi(\alpha)/s^2$, and

$$-V_\alpha - \frac{\psi(\beta)}{s^2} = \frac{\psi(\alpha)-\psi(\beta)}{s^2} - \frac{\psi'(\alpha)(\alpha-\beta)}{s(1+2\alpha)} \ge \psi'(\alpha)(\alpha-\beta)\Big[\frac{1}{s^2} - \frac{1}{s(1+2\alpha)}\Big] \ge 0$$

by concavity and $s \le 1+2\alpha$. Hence along every step the charge is at most the decrease of V, on either side of the diagonal. At the end, $W(\alpha) > \psi(\alpha)/(1+2\alpha)$ because ψ is increasing, so for $\alpha \ge \beta$

$$V(\alpha,\beta) > \frac{\psi(\alpha)}{s} \ge \frac{\Phi(\alpha-\beta)}{s} = \delta\,\Phi(R_\infty).$$

Lemma 4.4 and the start $(\alpha,\beta) = (0,0)$ give $E_i/\delta < V(0,0) = W(0) = \int_1^\infty \Phi(u)u^{-2}\,du$. Since $\Phi'(u) = 2u\ln(1+u^{-2})/\ln 2$ and $\Phi(1) = 2$, integration by parts and the antiderivative $u\ln(1+u^{-2}) + 2\arctan u$ give

$$\int_1^\infty \frac{\Phi(u)}{u^2}\,du = 2 + \frac{2}{\ln 2}\int_1^\infty \ln(1+u^{-2})\,du = 2 + \frac{2}{\ln 2}\Big(\frac{\pi}{2} - \ln 2\Big) = \frac{\pi}{\ln 2}.$$

For sharpness take $q = 1-2^{-m}$ and $K = m2^m$ gates alternating AND, OR, AND, ...; at an AND the sibling is an OR of m fresh variables and at an OR an AND of m fresh variables, so every $p_l = q$. Then α and β differ by $O(1-q)$ relative to s, every step multiplies s by $1/q$, and s runs from 1 to $q^{-K} \approx e^m$. The charges are Riemann sums of $\int_1^{q^{-K}} \Phi(u)u^{-2}\,du$ with mesh tending to zero, and the terminal term is nonnegative, so the ratio tends to $\pi/\ln 2$ as $m \to \infty$. These formulas have $1 + m^2 2^m$ variables. □

The values for $m = 2, 4, \ldots, 14$ are 3.633, 4.326, 4.492, 4.525, 4.531, 4.5322, 4.5323, and the identity of Lemma 4.4 agrees with brute-force transforms of random guarded paths to 3×10^{-15}. Read-once formulas are known to satisfy FEI [19, 4, 18], and for the whole one-coordinate step $H(f) - \frac{1}{2}H(f_{i\to 1}) - \frac{1}{2}H(f_{i\to -1})$ every leaf of a read-once formula has ratio below $5 + \ln 4$ in nats [21]; the content of Theorem 4.3 is that the erasure ratio is bounded at every leaf, however small its influence, with the exact constant.

5. Selection rules that fail

All three constructions below use one device, a single point added to a monotone function, which is also the source of the example in [6, Proposition 5.1]. Let $k = 2m+1$, write Maj for majority on k variables with value -1 when true, and let $\hat{\text{Maj}}$ denote its Fourier coefficients. For odd $|T| = s = 2j+1$,

$$|\hat{\text{Maj}}(T)| = \frac{\binom{2m}{m}}{4^m} \cdot \frac{\binom{m}{j}}{\binom{2m}{2j}}$$

[17, Section 5.3]. From $\binom{2m}{m} \ge 4^m/(2m+1)$, $\binom{m}{j} \ge 2^{mh_2(j/m)}/(m+1)$ and $\binom{2m}{2j} \le 2^{2mh_2(j/m)}$,

$$2^k|\hat{\text{Maj}}(T)| \ge \frac{2^{m+1}}{(2m+1)(m+1)} \qquad (|T| \text{ odd}). \tag{3}$$

5.1. **The coordinate of largest influence.**

Proposition 5.1. *For every C there is a monotone Boolean function whose coordinate of largest influence is unique and has $E_i/\mathrm{Inf}_i > C$.*

Proof. Fix an m-subset A of $[k]$ and let $F(x,y) = \mathrm{Maj}(y) \vee (x \wedge \bigwedge_{j\in A} y_j)$. The only y at which x matters is $y = 1_A$, so $\mathrm{Inf}_x(F) = 2^{-k}$, $\mathrm{D}_x F = \pm\mathbf{1}[y = 1_A]$ and $q_T = 4^{-k}$ for all $T \subseteq [k]$. Moreover $\mathbb{E}_x F = \mathrm{Maj} - \mathrm{Maj}(1_A)\mathbf{1}[y = 1_A]$ differs from Maj in each coefficient by 2^{-k}. Since $m(a,b) \geq b\log_2(1 + a/b)$, (2) and (3) give, for m large,

$$\frac{E_x(F)}{\mathrm{Inf}_x(F)} \geq 2^{-k} \sum_{|T| \text{ odd}} \log_2\bigl(1 + (2^k \hat{F}(T))^2\bigr) \geq \log_2\Bigl(\frac{2^{m+1}}{(2m+1)(m+1)} - 1\Bigr) = \frac{k}{2} - O(\log k).$$

Now replace each y_j by a majority of b fresh variables, b odd. Because the inner function is balanced, the spectral sample of the composition is that of F with each block refined independently by $\hat{\mathrm{Maj}}_b^{\,2}$, which has total mass one on nonempty sets; by homogeneity of m this leaves E_x and Inf_x unchanged, while each new leaf has influence $\mathrm{Inf}_{y_j}(F)\binom{b-1}{(b-1)/2}2^{1-b} \to 0$. For b large x is the unique coordinate of largest influence, and the composition is monotone. □

For $k = 3, 5, \ldots, 17$ the ratio of x is 3.52, 4.24, 4.99, 5.78, 6.62, 7.48, 8.37, 9.27.

5.2. **Most coordinates hidden.**

Proposition 5.2. *For $k = 2m+1$ let*

$$F(y,z) = [\,|y| \geq m+1\,] \;\vee \bigvee_{|A|=m} \Bigl(z_A \wedge \bigwedge_{j\in A} y_j\Bigr),$$

a monotone function of $k + \binom{k}{m}$ variables. Each of the $\binom{k}{m}$ coordinates z_A has influence 2^{-k} and

$$\frac{E_{z_A}(F)}{\mathrm{Inf}_{z_A}(F)} = 2^{-k} \sum_{T\subseteq[k]} \Phi\bigl(2^k \hat{a}(T)\bigr) \geq \frac{k}{2} - O(\log k),$$

where $a = \mathrm{Maj} - \mathbf{1}[|y| = m]$ and $\hat{a}(T) = \bigl(1 - \frac{s}{k+1}\bigr)\hat{\mathrm{Maj}}(T)$ for odd $|T| = s$.

Proof. On the layer $|y| = m$ the output is z_y, below it false and above it true. So $F = a(y) \pm \sum_A z_A \mathbf{1}[y = 1_A]$ with no term containing two hidden variables, and for z_A one has $q_T = 4^{-k}$ and $p_T = \hat{a}(T)^2$ on $T \subseteq [k]$, while frequencies containing another z_B have $q = 0$. This gives the identity. For the coefficients, let $\mathcal{L}$ be the Laplacian, $\mathcal{L}\chi_T = |T|\chi_T$. Majority changes only between the layers m and $m+1$, across $m+1$ edges from each point, so $\mathcal{L}\,\mathrm{Maj} = (m+1)(L(y) - L(\bar{y}))$ with $L = \mathbf{1}[|y| = m]$. At odd T this reads $s\,\hat{\mathrm{Maj}}(T) = (k+1)\hat{L}(T)$. The lower bound follows from (3), $1 - s/(k+1) \geq 1/(k+1)$ and $\Phi(u) \geq \log_2(1 + u^2)$, as before. □

So all but k of the $k + \binom{k}{m}$ coordinates have unbounded ratio. The k base coordinates stay good. With $r = \binom{k}{m}$ and $\delta = 2^{-k}$, a base coordinate y_1 has influence $r\delta$. A frequency containing one hidden variable z_B has coefficient $\pm 2^{-k}\chi_T(1_B)$ at $T \cup \{z_B\}$, $T \subseteq [k]$, so the pairs T, $T \cup \{1\}$ have equal squares 4^{-k}, and since $m(c,c) = 2c$ these frequencies contribute exactly $r\delta$ to E_{y_1}. The remaining frequencies give $E_{y_1}(a)$, so the ratio of y_1 is $1 + E_{y_1}(a)/(r\delta)$, which is 2.14 at $k = 17$. At $k = 17$ the hidden ratio is 14.45, against $\log_2\binom{17}{8} = 14.57$.

5.3. **Power weights.** Averaging E_i/Inf_i with weights Inf_i^{a+1} would give a good coordinate if

$$\sum_i \mathrm{Inf}_i(f)^a E_i(f) \le C \sum_i \mathrm{Inf}_i(f)^{a+1} \tag{4}$$

held for all monotone f. For $a = 0$ this is the monotone case of the FEI conjecture with $\sum_i E_i$ in place of H.

Proposition 5.3. *For every $a \neq 0$ there is no C for which (4) holds for all monotone functions.*

Proof. For $a < 0$ take F of Proposition 5.2, with $r = \binom{k}{m}$ and $\delta = 2^{-k}$. The hidden coordinates contribute $r\delta^{a+1}$ to the right side and the base coordinates $k(r\delta)^{a+1}$, a fraction $kr^a \to 0$ of it. The left side is at least $r\delta^a E_{z_A}$, so the quotient is at least $(E_{z_A}/\delta)/(1 + kr^a) \to \infty$. For $a > 0$ let $F(x, y) = \mathrm{Maj}(y) \vee (x \wedge \bigwedge_{j\in A} y_j)$ be the function in the proof of Proposition 5.1, before the majority substitution, and replace each y_j by a copy of the tribes function $T(w, s)$ on fresh variables, with $s = \mathrm{round}(2^w \ln 2)$. Call the result F_w; it is monotone. The tribes function has mean $\mu_w = 2(1 - 2^{-w})^s - 1$, and $\mu_w \to 0$ as $w \to \infty$. The expansion of Section 2.1 applies with x kept as a block of its own, expanded in the uniform basis, and each y_j expanded in the μ_w-biased basis. Whether $x \in U$ depends only on the first stage of the draw, so by homogeneity of m the quantities $E_x(F_w)$ and $\mathrm{Inf}_x(F_w)$ are those of F computed in this mixed basis. They are continuous in μ_w, and therefore converge to $E_x(F)$ and $\mathrm{Inf}_x(F) = 2^{-k}$. A leaf v in block j is pivotal for F_w exactly when it is pivotal for its copy of $T(w, s)$ and y_j is pivotal for F, and the two events are independent. Hence $\mathrm{Inf}_v(F_w) \le \mathrm{Inf}_v(T(w, s)) = 2^{1-w}(1 - 2^{-w})^{s-1} \le 2^{1-w}$. There are $kws < kw2^w$ leaves, so they contribute at most $kw2^w \cdot 2^{(1-w)(1+a)} = 2^{1+a}kw2^{-aw}$ to the right side of (4), and this tends to 0. The left side is at least $\mathrm{Inf}_x(F_w)^a E_x(F_w)$, since every term is nonnegative. Therefore the quotient of the two sides tends to at least $E_x(F)/\mathrm{Inf}_x(F)$ as $w \to \infty$, and by Proposition 5.1 this is unbounded in k. □

The proof for $a > 0$ fails at $a = 0$, where the leaves contribute $kws2^{1-w}(1-2^{-w})^{s-1} \sim kw \ln 2$ to the right side. Majority blocks, as in Proposition 5.1, only reach $a > 1$, because the leaf influences of Maj_b sum to order $\sqrt{b}$ while those of $T(w, s)$ sum to order w. The proposition excludes uniform averaging of the ratios ($a = -1$) and inverse-influence weights. Among power weights only $a = 0$ remains, and that case is the monotone conjecture in the form $\sum_i E_i \le C\, I$. The same construction excludes the weights $\mathrm{Inf}_i\, g(\mathrm{Inf}_i)$ for every g with $g(t)\log(1/t) \to 0$ as $t \to 0$: the leaf influences of F_w are of order 2^{-w}, so the leaves contribute $O(kw\, g(2^{1-w}))$ to the right side. It says nothing about weights at or above $\mathrm{Inf}_i/\log(1/\mathrm{Inf}_i)$, the scale of Talagrand's inequality [24], which keeps $\sum_i \mathrm{Inf}_i/\log(1/\mathrm{Inf}_i)$ at least a constant times the variance.

Weights that vary slowly give nothing easier. Let $g > 0$ be slowly varying at 0, that is $g(ct)/g(t) \to 1$ as $t \to 0$ for every $c > 0$ [2]; this includes the constants and $\log(1/t)^{-b}$ for every b. If $\sum_i g(\mathrm{Inf}_i(f))E_i(f) \le C \sum_i g(\mathrm{Inf}_i(f))\mathrm{Inf}_i(f)$ for all monotone f, then $\sum_i E_i(\psi) \le C\, I(\psi)$ for every balanced monotone ψ. To see this, let b be odd and $h = \mathrm{Maj}_b(\psi(x^1), \ldots, \psi(x^b))$ on disjoint blocks. Since $\hat{\psi}(\emptyset) = 0$, $\hat{h}$ at a frequency with nonempty parts S_j, $j \in T$, is $\widehat{\mathrm{Maj}}_b(T) \prod_{j\in T} \hat{\psi}(S_j)$. For a leaf i of block j, the pairs $(S, S \cup \{i\})$ with $S \neq \emptyset$ in block j keep T, and by homogeneity of m they give $\tau_b E_i(\psi)$, where $\tau_b = \binom{b-1}{(b-1)/2} 2^{1-b}$ is the influence of a coordinate of Maj_b. The pair $(\emptyset, \{i\})$ changes $|T|$ by one, so one of its two coefficients vanishes, because Maj_b has weight only on odd sets. Hence $E_{(j,i)}(h) = \tau_b E_i(\psi)$ and $\mathrm{Inf}_{(j,i)}(h) = \tau_b \mathrm{Inf}_i(\psi)$, and

$$\frac{\sum_v g(\mathrm{Inf}_v(h))E_v(h)}{\sum_v g(\mathrm{Inf}_v(h))\mathrm{Inf}_v(h)} = \frac{\sum_i g(\tau_b \mathrm{Inf}_i(\psi))E_i(\psi)}{\sum_i g(\tau_b \mathrm{Inf}_i(\psi))\mathrm{Inf}_i(\psi)} \longrightarrow \frac{\sum_i E_i(\psi)}{I(\psi)}$$

as $b \to \infty$, since $g(\tau_b \mathrm{Inf}_i)/g(\tau_b) \to 1$ for each of the finitely many i with $\mathrm{Inf}_i(\psi) > 0$; and h is monotone. The convergence is slow. For $\psi = x_1x_2 \vee x_2x_3 \vee x_1x_4$ and $g(t) = 1/\ln(1/t)$ the

weighted quotient is 1.1610 for ψ, 1.2012 at $b = 3$, 1.2346 at $b = 101$ and 1.2533 at $b = 10^6 + 1$, against $\sum_i E_i(\psi)/I(\psi) = 1.2683$. The argument needs $\hat{\psi}(\emptyset) = 0$. For biased ψ the pair $(\emptyset, \{i\})$ of ψ is not carried into h, and Maj_b in the biased basis has weight on even sets, so the two sides no longer scale together.

Remark 5.4. The spectral sample of a monotone function need not be positively correlated. For $x_1 \wedge (x_2 \vee x_3)$ one finds $\Pr[1 \in U] = 3/4$, $\Pr[2 \in U] = 1/4$ and $\Pr[\{1, 2\} \subseteq U] = 1/8 < 3/16$.

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics, and its results were developed with the Brain [3] and Math Brain [13]. In November 2026 a Claude Fable agent (Anthropic), connected to the Brain, studied the one-coordinate constant and its composition towers and directed Codex agents running OpenAI's GPT-6.1 on the same problem. The Codex agents ran the tower searches and the exhaustive search over $K_{3,3}$-invariant bases, and proposed the width bound, the read-once bound and the negative examples of Sections 4 and 5. A second Claude agent found the tower of Table 1 with the largest value and the pinned-mean limit, and a third gave the tribes substitution that extends the power-weight result to $0 < a \leq 1$ and the majority lift for slowly varying weights. Claude Opus 5.5 (Anthropic) searched the literature, wrote the proofs in their present form, recomputed every tower in Table 1 and checked every numerical claim with independent code, prepared the manuscript and audited it. These checks are not peer review. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Code and data availability

`https://github.com/yspennstate/fei-good-coordinates`

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [13].

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